



ECx00G&EC600U&EG800G Series

QuecOpen SMS API Reference Manual

LTE Standard Module Series

Version: 1.0

Date: 2023-09-04

Status: Released



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About the Document

Revision History

Version	Date	Author	Description
-	2023-05-25	Marvin NING	Creation of the document
1.0	2023-09-04	Marvin NING	First official release

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1 Introduction

Quectel ECx00G family, EC600U and EG800G series modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS system, which is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]**.

This document mainly introduces the SMS APIs of ECx00G family, EC600U and EG800G series modules and related examples in QuecOpen® solution. SMS APIs in this document can mainly realize features listed below:

- Send SMS messages
- Receive SMS messages
- Delete SMS messages
- Get SMS center number
- Get the storage status of SMS messages

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC600U Series
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
	EG800G Series

2 SMS APIs

2.1. Header File

ql_api_sms.h, the header file of SMS APIs, is located in the directory of *ql-sdk\components\ql-kernel\inc* in SDK. Unless otherwise specified, the header files mentioned in this document are all located in this directory.

2.2. API Functions Overview

Table 2: API Functions Overview

Function	Description
<i>ql_sms_get_init_status()</i>	Gets the initialization status of SMS
<i>ql_sms_send_msg()</i>	Sends SMS message in text format
<i>ql_sms_send_pdu()</i>	Sends SMS message in PDU format
<i>ql_sms_read_msg()</i>	Reads SMS message in text or PDU format
<i>ql_sms_read_msg_list()</i>	Reads the list of SMS messages
<i>ql_sms_delete_msg()</i>	Deletes SMS messages
<i>ql_sms_get_center_address()</i>	Gets the SMS center number
<i>ql_sms_set_center_address()</i>	Sets the SMS center number
<i>ql_sms_set_storage()</i>	Sets the storage location of SMS message
<i>ql_sms_get_storage()</i>	Gets storage location of SMS message
<i>ql_sms_get_storage_info()</i>	Gets the storage status of SMS message in (U)SIM card and ME
<i>ql_sms_callback_register()</i>	Registers the SMS message callback function

2.3. API Functions Description

2.3.1. ql_sms_get_init_status

This function gets the initialization status of SMS. The SMS related functions can only be called after initialization.

● Prototype

```
ql_sms_errcode_e ql_sms_get_init_status(uint8_t nSim, uint8_t *status)
```

● Parameter

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

status:

[Out] The initialization status of SMS.

0 Uninitialized

1 Initialized

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.1.1. ql_sms_errcode_e

The enumeration of SMS APIs execution error codes is defined as follows:

```
typedef enum
{
    QL_SMS_SUCCESS                = 0,
    QL_SMS_ERROR                  = 1 | (QL_COMPONENT_SMS << 16),
    QL_SMS_NOT_INIT_ERR,
    QL_SMS_PARA_ERR,
    QL_SMS_NO_MEMORY_ERR,
    QL_SMS_SEM_CREATE_ERR,
    QL_SMS_SEM_TIMEOUT_ERR,
    QL_SMS_NO_MSG_ERR,
}ql_sms_errcode_e
```

- **Parameter**

Parameter	Description
<i>QL_SMS_SUCCESS</i>	Successful execution
<i>QL_SMS_ERROR</i>	Failed execution
<i>QL_SMS_NOT_INIT_ERR</i>	SMS service is not initialized
<i>QL_SMS_PARA_ERR</i>	Parameter error
<i>QL_SMS_NO_MEMORY_ERR</i>	No memory
<i>QL_SMS_SEM_CREATE_ERR</i>	Fail to create semaphore
<i>QL_SMS_SEM_TIMEOUT_ERR</i>	Semaphore times out
<i>QL_SMS_NO_MSG_ERR</i>	No SMS messages to read

2.3.2. ql_sms_send_msg

This function sends SMS messages in text format.

- **Prototype**

```
ql_sms_errcode_e ql_sms_send_msg(uint8_t nSim, char *phone_num, char *data, ql_sms_code_e code)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

phone_num:

[In] The recipient's mobile phone number.

data:

[In] Message content. Length of the message content is less than 140 bytes.

code:

[In] SMS encoding method. See **Chapter 2.3.2.1** for details.

- 0 GSM-7 bit
- 1 UCS2

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.2.1. ql_sms_code_e

The SMS encoding method is defined as follows:

```
typedef enum
{
    GSM = 0,
    UCS2 = 1,
}ql_sms_code_e
```

- **Parameter**

Parameter	Description
<i>GSM</i>	GSM encoding
<i>UCS2</i>	UCS2 encoding, used when the sent message contains Chinese.

2.3.3. ql_sms_send_pdu

This function sends SMS messages in PDU format.

- **Prototype**

```
ql_sms_errcode_e ql_sms_send_pdu(uint8_t nSim, char *pdu)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

pdu:

[In] PDU message.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.4. ql_sms_read_msg

This function reads SMS messages in text or PDU format.

● Prototype

```
ql_sms_errcode_e ql_sms_read_msg(uint8_t nSim, uint8_t index, char *buf, uint16_t buf_len,
ql_sms_format_e format)
```

● Parameter

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

index:

[In] Index of SMS message.

buf:

[Out] An array pointer for receiving SMS message.

buf_len:

[In] The array length.

format:

[In] SMS message format. See **Chapter 2.3.4.1** for details.

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.4.1. ql_sms_format_e

The SMS message format is defined as follows:

```
typedef enum
{
    PDU = 0,
    TEXT = 1,
}ql_sms_format_e
```

- **Parameter**

Parameter	Description
<i>PDU</i>	PDU format
<i>TEXT</i>	Text format

2.3.5. ql_sms_read_msg_list

This function reads the list of SMS messages.

- **Prototype**

```
ql_sms_errcode_e ql_sms_read_msg_list(uint8_t nSim, ql_sms_format_e format)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

format:

[In] SMS message format. See **Chapter 2.3.4.1** for details.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.6. ql_sms_delete_msg

This function deletes SMS messages.

- **Prototype**

```
ql_sms_errcode_e ql_sms_delete_msg(uint8_t nSim, uint8_t index)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

index:

[In] The index of the SMS messages to be deleted.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.7. ql_sms_get_center_address

This function gets the SMS center number.

- **Prototype**

```
ql_sms_errcode_e ql_sms_get_center_address(uint8_t nSim, char* address, uint8_t len)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

address:

[Out] An array pointer for receiving the SMS center number.

len:

[In] The array length.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.8. ql_sms_set_center_address

This function sets the SMS center number. If there is no special requirement, it is not recommended to change the SMS center number.

● Prototype

```
ql_sms_errcode_e ql_sms_set_center_address(uint8_t nSim, char* address)
```

● Parameter

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

address:

[In] SMS center number in string format. The prefix "+86" is required.

● Return Value

See **Chapter 2.3.1.1** for details.

2.3.9. ql_sms_set_storage

This function sets the storage location of the SMS message.

● Prototype

```
ql_sms_errcode_e ql_sms_set_storage(uint8_t nSim, ql_sms_stor_e mem1, ql_sms_stor_e mem2, ql_sms_stor_e mem3)
```

● Parameter

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

mem1:

[In] The storage location for SMS messages that have been read and deleted. See **Chapter 2.3.9.1** for details.

mem2:

[In] The storage location for SMS messages to be written and sent. See **Chapter 2.3.9.1** for details.

mem3:

[In] The preferred storage location for SMS messages that have been received. See **Chapter 2.3.9.1** for details.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.9.1. ql_sms_stor_e

The enumeration of SMS message storage location is defined as follows:

```
typedef enum
{
    ME = 1,
    SM = 2,
}ql_sms_stor_e
```

- **Parameter**

Parameter	Description
<i>ME</i>	Mobile equipment (ME)
<i>SM</i>	(U)SIM card

2.3.10. ql_sms_get_storage

This function gets the storage location of SMS message.

- **Prototype**

```
ql_sms_errcode_e ql_sms_get_storage(uint8_t nSim, ql_sms_mem_info_t *mem_info)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

mem_info:

[Out] The storage location of SMS message. See **Chapter 2.3.10.1** for details.

- **Return Value**

See **Chapter 2.3.1.1** for details.

2.3.10.1. ql_sms_mem_info_t

The structure of SMS message storage location is defined as follows:

```
typedef struct
{
    ql_sms_stor_e mem1;
    ql_sms_stor_e mem2;
    ql_sms_stor_e mem3;
}ql_sms_mem_info_t
```

- **Parameter**

Type	Parameter	Description
<i>ql_sms_stor_e</i>	<i>mem1</i>	The storage location for SMS messages that have been read and deleted. See Chapter 2.3.9.1 for details.
<i>ql_sms_stor_e</i>	<i>mem2</i>	The storage location for SMS messages to be written and sent. See Chapter 2.3.9.1 for details.
<i>ql_sms_stor_e</i>	<i>mem3</i>	The preferred storage location for SMS messages that have been received. See Chapter 2.3.9.1 for details.

2.3.11. ql_sms_get_storage_info

This function gets the storage status of SMS message in the (U)SIM card and ME.

- **Prototype**

```
ql_sms_errcode_e ql_sms_get_storage_info(uint8_t nSim, ql_sms_stor_info_s *stor_info)
```

- **Parameter**

nSim:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

- 0 (U)SIM card 1
- 1 (U)SIM card 2

stor_info:

[Out] The structure of SMS message storage status. See **Chapter 2.3.11.1** for details.

- Return Value

See **Chapter 2.3.1.1** for details.

2.3.11.1.ql_sms_stor_info_s

The structure of SMS message storage status is defined as follows:

```
typedef struct
{
    uint16_t usedSlotSM;
    uint16_t totalSlotSM;
    uint16_t unReadRecordsSM;
    uint16_t usedSlotME;
    uint16_t totalSlotME;
    uint16_t unReadRecordsME;
    ql_sms_stor_e newSmsStorId;
}ql_sms_stor_info_s
```

- Parameter

Type	Parameter	Description
uint16_t	<i>usedSlotSM</i>	The number of SMS message stored in (U)SIM card (used storage memory)
uint16_t	<i>totalSlotSM</i>	The total number of SMS message that can be stored in (U)SIM card
uint16_t	<i>unReadRecordsSM</i>	The number of unread SMS message in (U)SIM card
uint16_t	<i>usedSlotME</i>	The number of SMS message stored in ME (used storage memory)
uint16_t	<i>totalSlotME</i>	The total number of SMS message that can be stored in ME
uint16_t	<i>unReadRecordsME</i>	The number of unread SMS message in ME
<i>ql_sms_stor_e</i>	<i>newSmsStorId</i>	The storage location of new SMS message. See Chapter 2.3.9.1 for details.

2.3.12. ql_sms_callback_register

This function registers the SMS message callback function.

- **Prototype**

```
void ql_sms_callback_register(ql_sms_event_handler_t cb)
```

- **Parameter**

cb:

[In] Callback function. See **Chapter 2.3.12.1** for details.

- **Return Value**

None

2.3.12.1. ql_sms_event_handler_t

This function defines the callback function for SMS message events.

- **Prototype**

```
typedef void (*ql_sms_event_handler_t)(uint8_t sim_id, int event_id, void *ctx)
```

- **Parameter**

sim_id:

[In] The (U)SIM card used. If the module only supports one (U)SIM interface, this parameter can only be set to 0.

0 (U)SIM card 1

1 (U)SIM card 2

event_id:

[In] SMS message events. See **Chapter 2.3.12.2** for details.

ctx:

[In] The pointer of inputting parameters for the callback function.

- **Return Value**

None

2.3.12.2.ql_sms_event_id_e

The enumeration of SMS message events is defined as follows:

```
typedef enum
{
    QL_SMS_INIT_OK_IND = 1 | (QL_COMPONENT_SMS << 16),
    QL_SMS_NEW_MSG_IND,
    QL_SMS_LIST_IND,
    QL_SMS_LIST_END_IND,
    QL_SMS_MEM_FULL_IND,
}ql_sms_event_id_e
```

● Parameter

Parameter	Description
<i>QL_SMS_INIT_OK_IND</i>	Successfully initialization of SMS
<i>QL_SMS_NEW_MSG_IND</i>	New SMS message
<i>QL_SMS_LIST_IND</i>	List of SMS message
<i>QL_SMS_LIST_END_IND</i>	The list of SMS message has been read
<i>QL_SMS_MEM_FULL_IND</i>	The storage memory of SMS message is full

3 SMS Demo

This chapter mainly introduces how to use the above APIs to develop SMS functions on the Application side.

3.1. Demo Introduction

sms_demo.c, provided in the SDK of Quectel ECx00G, EC600U and EG800G family modules, is located in the directory of *components\ql-application\sms*. The demo contains some functions including sending, reading and deleting SMS messages. The following example shows how to use the API to send English and Chinese SMS messages.

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#include "ql_api_common.h"
#include "ql_api_osi.h"
#include "ql_api_sms.h"
#include "ql_log.h"

ql_task_t sms_task = NULL;
ql_sem_t sms_init_sem = NULL;

void user_sms_event_callback(uint8_t nSim, int event_id, void *ctx)
{
    switch(event_id)
    {
        case QL_SMS_INIT_OK_IND:
        {
            QL_SMS_LOG("QL_SMS_INIT_OK_IND");
            ql_rtos_semaphore_release(sms_init_sem);
            break;
        }
        case QL_SMS_NEW_MSG_IND:
        {
            ql_sms_new_s *msg = (ql_sms_new_s *)ctx;
```

```

        QL_SMS_LOG("sim=%d, index=%d, storage memory=%d", nSim, msg->index,
msg->mem);
        break;
    }
    case QL_SMS_MEM_FULL_IND:
    {
        ql_sms_new_s *msg = (ql_sms_new_s *)ctx;
        QL_SMS_LOG("QL_SMS_MEM_FULL_IND sim=%d, memory=%d",nSim,msg->mem);
        break;
    }
    default :
        break;
}
}

void sms_demo_task(void * param)
{
    uint8_t nSim = 0;
    QL_SMS_LOG("enter");
    ql_sms_callback_register(user_sms_event_callback);

    //wait sms ok
    if(ql_rtos_semaphore_wait(sms_init_sem, QL_WAIT_FOREVER)){
        QL_SMS_LOG("Waiting for SMS init timeout");
    }

    //Send English text message
    if(QL_SMS_SUCCESS == ql_sms_send_msg(nSim,"+86189xxxxxxx","Hello,marvin", GSM)){
        QL_SMS_LOG("send sms OK");
    }else{
        QL_SMS_LOG("send sms FAIL");
    }
    //Send messages in Chinese and English. (Need use UTF8 encoding to open sms_demo.c for
chinese.)
    if(QL_SMS_SUCCESS == ql_sms_send_msg(nSim,"+86189xxxxxxx","hello,你好", UCS2)){
        QL_SMS_LOG("send pdu sms OK");
    }else{
        QL_SMS_LOG("send pdu sms FAIL");
    }
    ql_rtos_task_delete(NULL);
}

```

```

QIOSStatus ql_sms_app_init(void)
{
    QIOSStatus err = QL_OSI_SUCCESS;

    err = ql_rtos_task_create(&sms_task, 4096, APP_PRIORITY_NORMAL, "SMS_TASK",
    sms_demo_task, NULL, 2);
    if(err != QL_OSI_SUCCESS)
    {
        QL_SMS_LOG("sms_task created failed, ret = 0x%x", err);
    }

    err = ql_rtos_semaphore_create(&sms_init_sem, 0);
    if(err != QL_OSI_SUCCESS)
    {
        QL_SMS_LOG("sms_init_sem created failed, ret = 0x%x", err);
    }

    return err;
}
    
```

3.2. Demo Autorun

The Demo is not enabled by default in *ql_init_demo_thread()*. As shown in the figure below, uncomment the codes when you run a test. In addition, you need to change the target phone number of the *ql_sms_send_msg()* in the demo to the your own phone number.

```

:
:
: #ifdef QL_APP_FEATURE_SMS
:     //ql_sms_app_init();
: #endif
: #ifdef QL_APP_FEATURE_VOICE_CALL
:     //ql_voice_call_app_init();
: #endif
: #ifdef QL_APP_FEATURE_VOLTE
:     //ql_volte_app_init();
: #endif
:
:
    
```

4 Appendix References

Table 3: Related Document

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen_CSDK_Quick_Start_Guide

Table 4: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
App	Application
GSM	Global System for Mobile Communications
IoT	Internet of Things
LTE	Long-Term Evolution
ME	Mobile Equipment
PDU	Packet Data Unit
RTOS	Real-Time Operating System
SDK	Software Development Kit
(U)SIM	(Universal) Subscriber Identity Module
SMS	Short Message Service
UCS	Universal Character Set
UTF	Unicode Transformation Format